

Hrithik Yash

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EDUCATION

University of California, San Diego (UCSD)

La Jolla, CA

Bachelor of Science in Mechanical Engineering (GPA: 3.82 / 4.00)

Expected May 2027

- **UCSD Physical Sciences Cohort Scholar** – Selected as one of 35 students campus-wide.
- **TRELS Scholar (Spring 2025)** – UCSD Undergraduate Research Fellowship.
- **Teaching:** PHYS 2A-C (Introductory Physics), MATH 18 (Linear Algebra), MATH 20E (Analytical Calculus).
- **7X Provost Honours**

RESEARCH AND ACADEMIC EXPERIENCE

Safe and Autonomous Systems (SAS) Lab, UC San Diego

La Jolla, CA

Research Assistant – Prof. Sylvia Herbert

June 2025 – Present

- Led a joint SAS Lab–Active Matter Lab effort enabling autonomous Crazyflie landing on a self-assembled MASBot platform.
- Implemented a ROS 2 control pipeline for Crazyflie drones, integrating state estimation, OptiTrack motion capture, and real-time command interfaces.
- Implemented an LQR controller for Crazyflie flight stabilization and precision landing, validated in simulation and on hardware.
- Experimentally calibrated MASBot inertial mass ($m \approx 0.25$ kg) and linear drag coefficient ($\gamma \approx 0.10$ kg/s) from free-decay trials by minimizing multi-step trajectory reconstruction error under an inertial ODE model.
- Formulated and compared competing dynamical hypotheses (overdamped vs inertial, pairwise vs wall-mediated interactions) using forward simulation and cross-trial validation to determine structurally necessary model components.
- Developed a non-parametric, distance-binned regression framework to learn interaction force basis functions directly from data; estimated coefficients via ridge-regularized least squares and validated learned kernels through forward simulation.

Active Matter Lab, UC San Diego

La Jolla, CA

Research Assistant – Prof. Tzer Han Tan

Sept. 2024 – Present

- Led a TRELS-funded research project on synchronization in Magnetically Augmented Spinning Robot Swarms (MASBots), presented at the Undergraduate research conference.
- Modeled MASBots as physically coupled nonlinear oscillators and experimentally validated phase-locking and synchronization dynamics.
- Redesigned MASBots using Fusion 360 to be 40% lighter, 60% more stable, and 25% cheaper through geometry and material optimization.
- Demonstrated that spatial confinement expands the synchronization window by 6×10^6 , enabling tunable collective coordination.

IIT Madras – Civil Engineering Department

Chennai, India

Research Intern

June – Aug. 2024

- Conducted Digital Image Correlation (DIC) analysis on textile-reinforced concrete, improving strain measurement resolution by 20%.
- Modeled test specimens and fixtures in Fusion 360 and AutoCAD for mechanical testing.
- Produced publication-quality visualizations and optimized fixture geometry for tensile tests.

Triton Droids – Robotics Club, UC San Diego

San Diego, CA

Member / Project Contributor

Sept. 2024 – May 2025

- Developed Matlab scripts to simulate Bi-Pedal motion
- Implemented a Python-based tool to parse Onshape geometry and generate optimized URDFs for MuJoCo environments
- Represented Triton Droids at automation conferences, successfully networking with industry leaders to secure corporate sponsorships

PROJECTS

Triton Racing – Formula SAE

Fall 2024

- Modeled key racecar components in SolidWorks, including front anti-roll bar, brake bobbin, and chassis tab.
- Performed Finite Element Analysis (FEA) on a custom wheel stand, improving the factor of safety from 0.05 to 2.0 through iterative design.
- Selected AISI 4130 steel to optimize strength-to-weight ratio and manufacturability.
- Ensured compliance with Formula SAE regulations, including aerodynamic and brake system standards.

Introduction to Mechanical Design

Spring 2024

- Built a fully functional competition-grade shooting robot that collected and shot balls .
- Utilized 3D printing and laser cutting for rapid prototyping and assembly.
- Designed and optimized a flywheel mechanism that improved points scoring by 40 percent
- Designed and assembled a fully functional pendulum clock in acrylic using Lasercamm.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Java, MySQL, HTML/CSS, JavaScript

Frameworks & Tools: ROS 2, Linux, Git, Docker, AprilTag, OptiTrack

CAD & Simulation: Fusion 360, SolidWorks, AutoCAD, MuJoCo, Ansys

Other: Data Visualization, Figma, Canva, Google Workspace